

Workers' Compensation Loss Reserving

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Valuation: December 31, 2007

Results and scope

The manual and package chain-ladder calculations both estimate **\$381.599 million** of unpaid losses through development age 10. The original reserve discrepancy is fully reconciled. The two BF calculations below show how the result changes when the expected loss ratio (ELR) changes. Neither ELR is an independent prior.

Calculation	Unpaid (USD millions)
Chain-ladder / Mack (common basis)	381.599
BF: mature-year paid ELR sensitivity	750.247
BF: CL-implied ELR sensitivity	471.488

Mack's standard error is **\$47.955 million**, with a coefficient of variation of **12.57%**. These apply to the common-basis estimate above. They are conditional on the model and exclude development beyond age 10.

The book and valuation

The source is the CAS Schedule P workers' compensation dataset. Code 388 is labelled **Federal Ins Co Grp**, with **Single** = 0, identifying an insurer group. The actual file has 90 records for this group, covering accident years 1998-2006 and ten development lags. Accident year 2007 is absent for this group; no row was invented to fill it.

Only observations with accident year + development lag - 1 at or before 2007 enter the fitted triangle. There are 54 such observations; 36 later observations are excluded. This is a historical portfolio study, not a current reserve recommendation.

What the estimates mean

The paid-loss measure includes allocated expenses and is net of reinsurance. Net earned premium is used for the ELR calculations. All source monetary values are interpreted as USD thousands; report and chart totals use USD millions. The estimates cover future payments on reported and unreported claims. This is **unpaid losses**, not solely incurred-but-not-reported claims. The package's **IBNR** output field is used as an unpaid-loss quantity in this paid triangle.

"Ultimate" in the calculation means projected cumulative paid at age 10. No later development is assumed. It does not establish that all claims are settled.

Why the original reserves differed

The manual calculation used development ages 1-10. The old Mack input used ages 1-9, dropping the sole observed age-10 payment. This changed both the projection horizon and the latest paid amount for accident year 1998.

Basis	Latest paid	Projected paid	Unpaid
Original Mack: ages 1-9	1,799.432	2,148.649	349.217
Manual CL: ages 1-10	1,803.472	2,185.071	381.599
Mack: ages 2-10	1,803.472	2,185.071	381.599

All amounts above are USD millions. The reduction in projected paid losses was \$36.423 million, partly offset by a \$4.040 million reduction in latest paid. Since unpaid = projected paid - latest paid, the net difference is **\$32.383 million**, or **8.49%** of the manual estimate. It was not immaterial.

The final observed development factor

The age-9-to-10 factor is 1.016951428, based only on the 1998 pair: 242,368 / 238,328 in USD thousands. Dropping it removes a roughly 1.70% multiplier from the age-9 projected totals. Its impact accumulates across accident years.

Accident year	Latest paid change	Projected change	Unpaid change
1998	4.040	4.040	0.000
1999	0.000	4.267	4.267
2000	0.000	4.138	4.138
2001	0.000	3.205	3.205
2002	0.000	3.517	3.517
2003	0.000	3.816	3.816
2004	0.000	4.334	4.334
2005	0.000	4.505	4.505
2006	0.000	4.601	4.601

Changes are full-horizon minus original trimmed estimates, in USD millions.

Common-basis comparison

The installed ChainLadder version is 0.2.21. Its standard Mack input supports at least as many origin rows as development columns; a square shape is not universally required. The original 9-by-10 input is rejected directly. The documentation also describes other rectangular cases.

For this book, using ages **2-10** gives a standard 9-by-9 triangle. Every origin has already reached age 2, so the unused age-1-to-2 factor is omitted while all latest payments and needed factors are retained. Volume weighting (**alpha** = 1), accident years, and the no-tail assumption are unchanged. Manual and package latest paid, projected paid and unpaid losses agree by origin within **1e-7** in USD thousands. No unexplained reserve difference remains.

Bornhuetter-Ferguson sensitivities

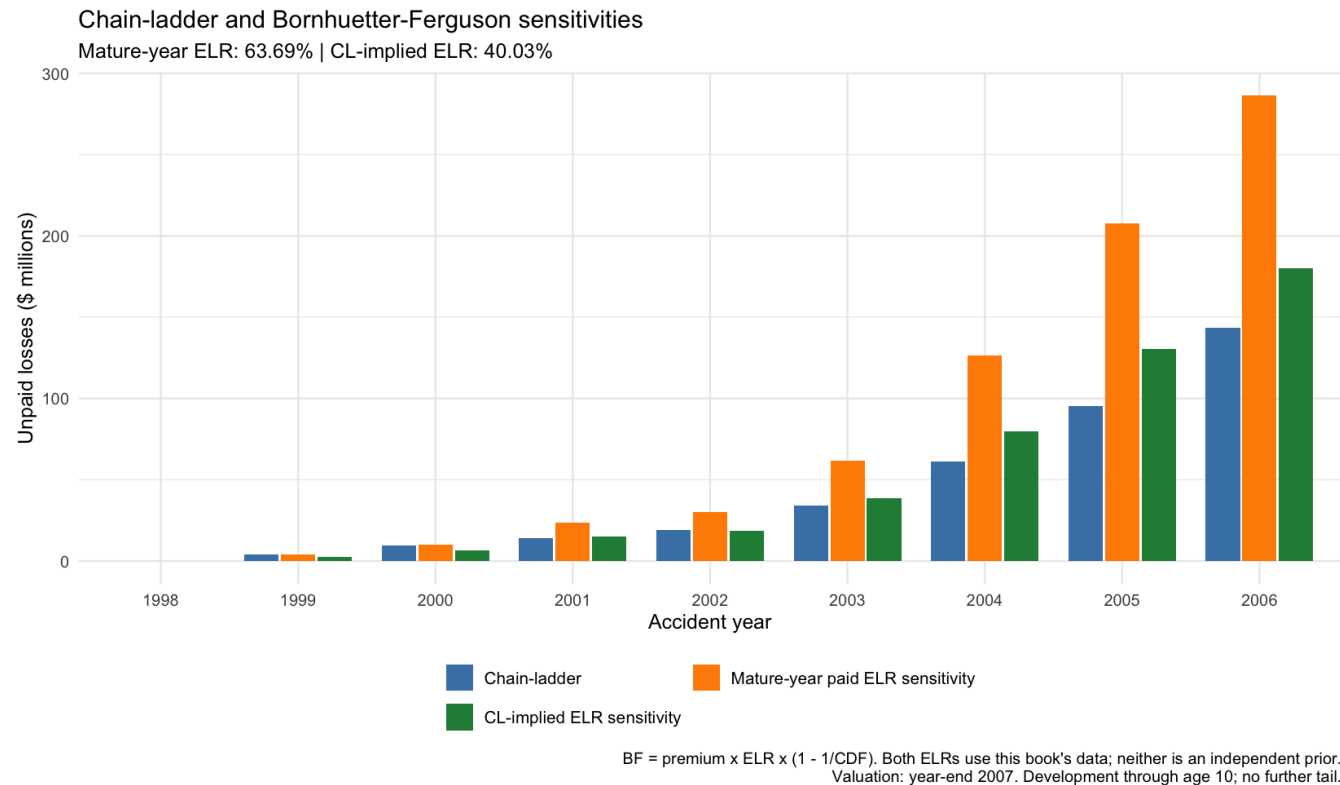
For paid-loss development, the BF estimate is:

$$\text{Unpaid}_i = \text{Premium}_i \times \text{ELR} \times (1 - 1/\text{CDF}_i).$$

The original ELR uses observed paid losses divided by net earned premium for years with CDF below 1.05. Those years are **1998, 1999, 2000**, not just 1998-1999. This gives **63.692681%**. A low CDF does not prove settlement or make this an external expectation.

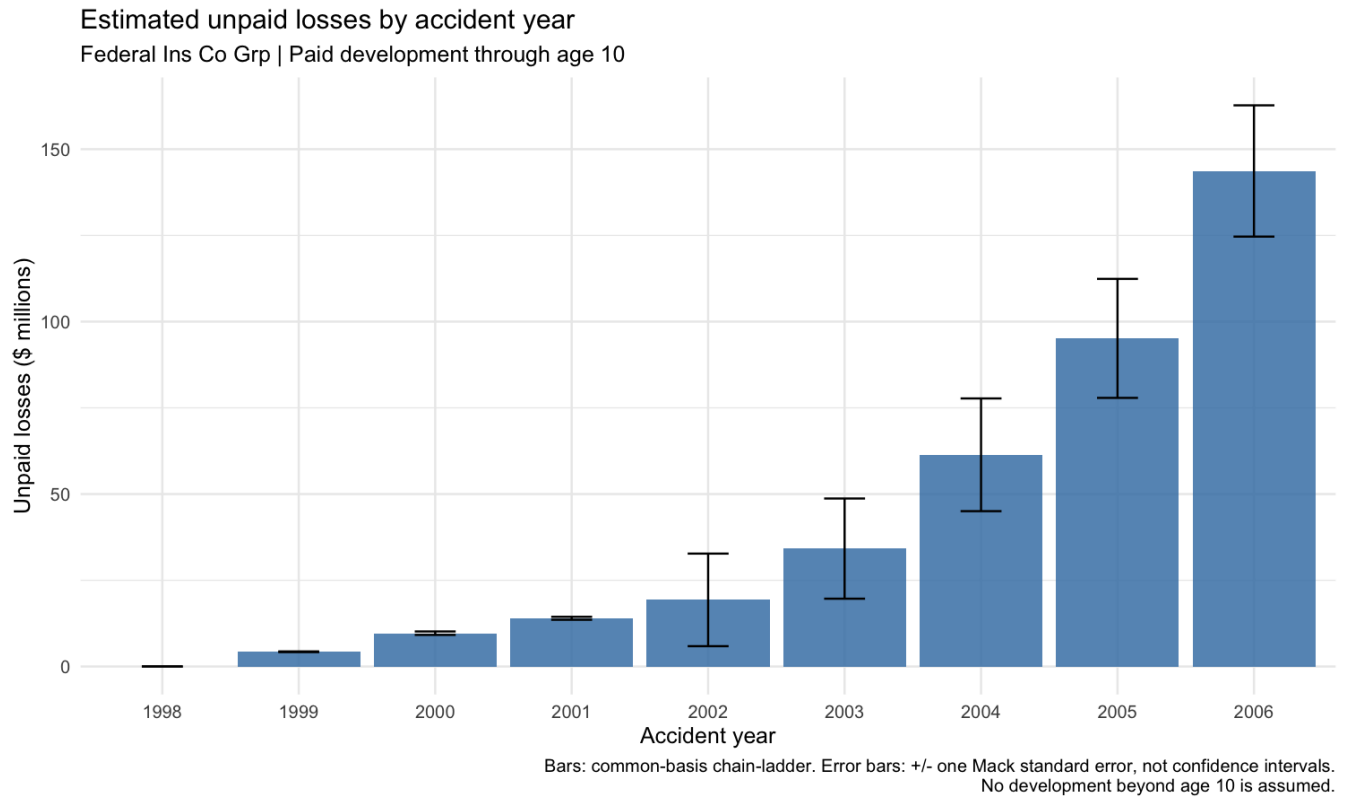
The second scenario uses total CL projected losses divided by total net earned premium: **40.027289%**. It is labelled **CL-implied ELR sensitivity**. The computation uses full precision; 40.0% was a rounded display label.

Scenario	ELR	Unpaid (USD millions)
Mature-year paid ELR	63.692681%	750.247
CL-implied ELR	40.027289%	471.488
Exactly 40% (rounding check)	40.000000%	471.167



Interpretation. The difference follows from the ELR inputs applied to the same premiums and unpaid proportions. The CL-implied scenario reuses CL results, so its proximity to CL is not independent validation. Neither scenario is labelled “correct,” “conservative,” or a preferred reserve based only on proximity.

Uncertainty on the common basis



Accident year	Latest age	CDF	Unpaid (millions)	SE (millions)
1998	10	1.000	0.000	0.000
1999	9	1.017	4.267	0.103
2000	8	1.040	9.657	0.500
2001	7	1.078	13.975	0.429
2002	6	1.101	19.312	13.423
2003	5	1.176	34.184	14.525
2004	4	1.309	61.382	16.343
2005	3	1.543	95.139	17.254
2006	2	2.086	143.683	19.032

Mack standard errors estimate prediction uncertainty under chain-ladder conditional mean and variance assumptions and independence between accident years. They include process and parameter risk within the selected model. The total SE includes dependence from shared factor estimates; it is not the sum of the per-year SEs. The plotted one-SE bands are not confidence intervals.

The final development factor has only one observed pair, so the final variance uses the package's `est.sigma` = "Mack" approximation. Arithmetic agreement does not establish the model assumptions. The old roughly 16% CV described the trimmed-age-9 calculation; it cannot be attached to the age-10 reserve. The common-basis CV is 12.57%.

Data checks and limits

Check	Result
Input records	12100
Company records	90
Observed cells through 2007	54
Excluded later cells	36
Duplicate company/year/lag records	0
Missing required observed cells	0
Zero paired denominators	0
Negative observed paid increments	1
Mature years: $CDF < 1.05$	1998, 1999, 2000

The code checks required numeric fields and finite values, integer years and lags, unique company/year/lag keys, expected observed cells through the valuation date, and numeric column ordering. Each accident year's premium must agree across development lags. Direct and assumed premium less ceded premium must equal net earned premium. Non-positive paired development denominators stop the analysis for investigation instead of causing observations to be dropped.

The 2001 paid-loss decrease

Cumulative paid for accident year 2001 decreases from \$190.882 million at age 6 to \$178.283 million at age 7: a **\$12.599 million decrease**. The original “smooth development” description was unsupported.

The supplied file matches the official CAS download byte for byte, and the decrease is present in the official records. The CSV does not explain its cause or provide transaction details to resolve it. It is retained in the factors and flagged in every run. No recovery, correction or other economic explanation is asserted without evidence. Resolving the cause would require underlying filings or clarification from the data provider.

Limits of the analysis

- **Tail:** projected paid at age 10 is treated as ultimate. The dataset does not establish that payments beyond this horizon are zero. The Mack SE excludes uncertainty about omitted later development.
- **Model:** stability of development and independence across origin years have not been established by the synthetic tests. Calendar effects, business mix, reinsurance changes and other model misspecification are not quantified here.
- **ELRs:** both BF scenarios use this book's own experience. Neither supplies independent evidence about expected losses.
- **Coverage:** the selected group's 2007 accident year is not present. The supplied posted-reserve field is not used as a benchmark because its coverage is not established as matching this paid-loss projection.
- **Validation:** later observations exist, but no runoff backtest is implemented. No predictive validation, credibility ranking or reserve recommendation is claimed.

Reproduction and sources

From the repository root, run:

```
Rscript build.R
```

The RStudio Console equivalent is `source("build.R")`. It launches a fresh R process, restores the recorded package environment, runs 16 synthetic test groups, and generates the CSV tables, three figures, README and this PDF. Both document sources read the same objects generated by the current analysis. No headline result is maintained separately by hand.

Environment and files

The recorded R version is 4.5.2. ChainLadder is 0.2.21; renv is 1.2.4. The lockfile records package versions and sources. Pandoc and a LaTeX distribution are external prerequisites; renv does not install or pin their complete system environments. Local build versions and session details are saved under `outputs/`.

The editable sources are `report/report.Rmd` and `report/README.Rmd`. The shared calculation functions are in `R/reserving.R`; the main analysis is `wc_loss_reserving.R`. Tests use synthetic data and can run without the CAS CSV. Tables and detailed input checks are regenerated under `outputs/tables/`, which is kept local. Only the intended report, charts, source and environment files are prepared for repository review.

Data acquisition

Download `wkcomp_pos_98-07.csv` from the CAS page's current workers' compensation link and place it at the repository root. The earlier 2011 release is a different file and schema. Read the usage terms and attribution requirements. The source CSV is not redistributed. The build checks its SHA-256 against the reference release so a changed input cannot silently reuse the report's scope statements.

Reference SHA-256:

```
8d0b02bed0939e932f9078f65266e9a3
98f580f90e5227cde053dd5b520affef
```

Sources

1. CAS loss-reserving data and field descriptions. Data accessed from this CAS page as compiled for CAS by S&P Global Market Intelligence. Official and local CSV equality checked September 4, 2026.
2. Official workers' compensation CSV.
3. CAS terms of use.
4. ChainLadder: MackChainLadder documentation. Supported shapes, weighting, standard errors and tail settings.
5. renv: restoring a project environment.